Parallel Isolation and Selective Expansion of Primary Porcine Retinal Endothelial Cells and Pericytes from a Single Donor Source

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Background/intro:
Retinal endothelial cells and pericytes constitute the cellular core of the inner blood-retina barrier and are central to vascular stability and barrier function. Experimental investigation of their coordinated responses requires primary cultures that retain physiological characteristics. This study aimed to establish a cost-effective and reproducible method for the parallel isolation and selective expansion of primary porcine retinal endothelial cells (pECs) and pericytes (pPCs) from the same retinal preparation, thereby providing a translationally relevant *in vitro* platform for studies of retinal vasculature.

Methods: Fresh porcine eyes obtained as an excess product from the food industry were disinfected and dissected. The retinal tissue was collected and subjected to enzymatic digestion to generate a mixed microvascular cell suspension. Selective culture conditions were applied to favour either endothelial or pericyte expansion. Cell morphology was assessed microscopically, and phenotypic identity was validated using RT-qPCR and immunofluorescent staining for cell-type specific markers.

Results: The protocol enabled reproducible establishment of enriched pEC and pPC-like monocultures. pEC cultures formed cobblestone monolayers and expressed endothelial markers at both transcript (PECAM1, vWF, ERG) and protein (VE-Cadherin, ERG) levels. pPC cultures exhibited spindle-shaped morphology and expressed mural-associated markers at RNA (PDGFRB, DES, NG2, PRKG1, ACTA2) and protein ($PDGFR_{\alpha/\beta}$, NG2, α-SMA) levels, indicative of a predominant pericyte phenotype. Markers of the alternate vascular cell type and glial cells were not detected, supporting selective enrichment of the intended populations.

Conclusions: The presented approach allows parallel isolation of pECs and pPCs from the same retinal source without reliance on advanced cell-sorting technologies. Cells isolated with the co-isolation protocol preserves key cellular characteristics of the retinal microvasculature and offers a practical platform for mechanistic studies of blood-retina barrier or retinal vascular diseases. To our knowledge, this is the first protocol presenting the successful isolation of porcine retinal microvascular cells.